

## TRAINING PHASE

Prior  
 $p(\theta)$

sample  $\theta$

Simulator  
 $p(Y|\theta)$

Simulate  $N$  times

Simulated  
Data  
 $\{(\theta_i, Y_i)\}_{i=1}^N$

Train Network

Neural  
Density  
Estimator  
 $q_\phi(\theta|Y)$

Evaluate

Posterior  
 $p(\theta|Y_{\text{obs}})$

## INFERENCE PHASE

Observed  
Data  
 $Y_{\text{obs}}$